Binary Tree Traversals (Inorder, Preorder, Postorder, Level Order) Data Structures

> **Definition: Tree Traversal** is the systematic process of visiting (reading, processing, or printing) every node in a tree data structure exactly once.

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1. Detailed Technical Explanation

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### 1. Depth-First Traversals (DFS):

1. **Preorder Traversal (Root -> Left -> Right):**

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### 2. Breadth-First Traversal (BFS / Level Order):

- Visits nodes level-by-level from top to bottom, and left to right at each level using a **FIFO Queue**.

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## 2. Complete Executable C Program for Traversals

```c

#incluc

struct Node {

int data

struct Node* createNode(int val) {

struct l

// 1. Inorder Traversal: Left -> Root -> Right

void in

// 2. Preorder Traversal: Root -> Left -> Right

void pr

// 3. Postorder Traversal: Left -> Right -> Root

void po

int main() {

// Cons

printf("Preorder : "); preorder(root); printf("\n");

printf("

---

## 3. Reconstructing Unique Binary Tree from Traversals

- A unique binary tree can be constructed if and only if **INORDER** is given along with either **PREORDER** or **POSTORDER**.

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## 4. Must-Write Points for Exams

- Inorder traversal of BST gives non-decreasing sorted order.

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## 5. Quick Recall Flow

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